

Sunan Zhao

PhD Student · Southern University of Science and Technology (SUSTech)

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Research Interests

AI for fluid mechanics, including physics-informed (PI) approaches and neural operator models.

Education

Ph.D. candidate in Mechanics (combined master's–doctoral program), Southern University of Science and Technology (SUSTech)

Sep 2023 – Present

Supervisor: Prof. Jianchun Wang

B.Eng. in Engineering Mechanics, China University of Petroleum (East China)

Sep 2019 – Jul 2023

Honors & Awards

Outstanding Graduate Student, Southern University of Science and Technology, 2024–2025 academic year

Publications

[1] Sunan Zhao, Zhijie Li, Boyu Fan, Yunpeng Wang, Huiyu Yang, Jianchun Wang*, "LESnets (Large-Eddy Simulation nets): Physics-informed neural operator for large-eddy simulation of turbulence," *Journal of Computational Physics*, vol. 537, 114125, 2025.

[2] Zhihong Guo, Sunan Zhao, Huiyu Yang, Yunpeng Wang, Jianchun Wang*, "Physics-Informed Transformer operator for the prediction of three-dimensional turbulence," *Acta Mechanica Sinica*, 2026.

[3] Sunan Zhao, Yunpeng Wang, Huiyu Yang, Zhihong Guo, Jianchun Wang*, "Large-eddy simulation nets (LESnets) based on physics-informed neural operator for wall-bounded turbulence," *arXiv preprint arXiv:2604.26621*, 2026.

Selected Presentations

LESnets: A physics-informed neural operator for three-dimensional large-eddy simulation of turbulence. *Chinese Congress of Theoretical and Applied Mechanics (CCTAM 2025)*, Changsha, China, 2025 (oral).

A physics-informed neural operator for wall-bounded turbulence prediction. *The 3rd Chinese Conference of Aerodynamics*, Shenyang, China, 2026 (oral).

A physics-informed neural operator for wall-bounded turbulence prediction. *The 14th National Conference on Fluid Mechanics*, Qingdao, China, 2026 (oral).